

Periodic Table of Elements vAsbestos

Super Seven

HI
HBr
HCl
HNO₃
H₂SO₄
HClO₃
HClO₄

Gas

NO
NO₂
CO
CO₂
CH₄
C₂H₆
C₃H₈
C₄H₁₀
N₂O
NH₃
SO₃
SO₂
H₂S
HCl

Conversions

1 L · atm = 101.3 J
K = °C + 273.15
°C = $\frac{5}{9}(\text{°F} - 32)$
°F = $\frac{9}{5}\text{°C} + 32$
1 cal = 4.184 J
1 lb = 453.59 g
1 atm = 760 mmHg
= 760 torr = 101.325 kPa
1 bar = 10⁵ Pa = 10⁵ N/m²

Periodic Trends

$\mathbb{Z}_{eff}$ increase →↓
EN, IE, & EA increase →↑
Radius & Metallic decrease →↑

Equilibrium

When $aA + bB \rightleftharpoons cC + dD$,
 $K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$
 $K_p = \frac{(P_C)^c(P_D)^d}{(P_A)^a(P_B)^b}$
 $K_a = \frac{[H^+][A^-]}{[HA]}$
 $K_b = \frac{[OH^-][HB^+]}{[B]}$
 $K_w = K_aK_b = [H^+][OH^-]$
 $K_w = 1.0 \times 10^{-14}$ (25° C)
pH = $pK_a + \log \frac{[A^-]}{[HA]} = -\log[H^+]$
pH + pOH = 14.
 $pK_a = -\log K_a, pK_b = -\log K_b.$

Gasses/Solutions

$PV = nRT$
 $P_A = P_{\text{total}}X_A$, where $X_A = \frac{\text{moles } A}{\text{total moles}}$
 $P_{\text{total}} = P_A + P_B + P_C + \dots$
 $M = \frac{\text{moles solute}}{\text{Liters solution}}, m = \frac{\text{moles solute}}{\text{kg solvent}}$
 $M_1V_1 = M_2V_2$ for dilution
STP = 273.15 K and 1.0 atm
At STP, ideal gas 22.4L mol⁻¹.
Standard conditions 25° C, 1 atm.
 $v_{rms} = \sqrt{\frac{3RT}{\mathcal{M}}}$

Quantum

$E_{\text{photon}} = hf = \frac{hc}{\lambda} \implies c = \lambda f$
 $\lambda = \frac{h}{mv}$
 $R_{H_{\text{Rydberg}}} = 1.097 \times 10^7 \text{ m}^{-1}$
 $\frac{1}{\lambda} = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$
 $\Delta E = (-2.18 \times 10^{-18} \text{ J}) \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$

Thermo/Electrochem

$q = mc\Delta T, \Delta E = q + w, H = E + PV$
 $\Delta S^\circ = \sum_{\text{products}} S^\circ - \sum_{\text{reactants}} S^\circ$
↑ Likewise for ΔH° and ΔG° ↑
 $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$
 $= -RT \ln K = -nFE^\circ$
 $I = \frac{q}{t}$
 $E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q.$

Kinetics

$[A]_t - [A]_0 = -kt$ (0th order)
 $\ln[A]_t - \ln[A]_0 = -kt$ (1st order)
 $\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$ (2nd order)
 $t_{1/2} = \frac{0.693}{k}$ (1st order)

Solutions (cont.)

$\Delta T_b = K_bmi \quad \Delta T_f = -K_fm_i$
 $P_A = X_AP_A^\circ \quad S_g = kP_g$
 $\Pi = \left(\frac{n}{V} \right) RT = MRT$
 $(K_{b_{\text{water}}}, K_{f_{\text{water}}}) = (0.512, 1.86)^\circ\text{C/m}$
 $\Delta H_{\text{fus water}} = 6.008 \text{ kJ/mol}$
 $\Delta H_{\text{vap water, 100}^\circ\text{C}} = 40.67 \text{ kJ/mol}$
 $c_{\text{ice}} = 2.093 \text{ J/(}^\circ\text{C)}$
 $c_{\text{water}} = 4.184 \text{ J/(}^\circ\text{C)}$
 $c_{\text{steam}} = 1.841 \text{ J/(}^\circ\text{C)}$

18 VIIIA

2
He
Helium
4.00

13 IIIA

5 2.04
B
Boron
10.81

14 IVA

6 2.55
C
Carbon
12.01

15 VA

7 3.14
N₂
Nitrogen
14.01

16 VIA

8 3.44
O₂
Oxygen
16.00

17 VIIA

9 3.98
F₂
Fluorine
19.00

10
Ne
Neon
20.18

3

11 0.93
Na
Sodium
22.99

2 IIA

4 1.57
Be
Beryllium
9.01

12 1.31
Mg
Magnesium
24.31

4

19 0.82
K
Potassium
39.10

20 1.00
Ca
Calcium
40.08

21 1.36
Sc
Scandium
44.96

22 1.54
Ti
Titanium
47.87

23 1.63
V
Vanadium
50.94

24 1.66
Cr★
Chromium
52.00

25 1.55
Mn
Manganese
54.94

26 1.83
Fe
Iron
55.85

27 1.88
Co
Cobalt
58.93

28 1.91
Ni
Nickel
58.69

29 1.90
Cu★
Copper
63.55

30 1.65
Zn
Zinc⁽²⁺⁾
65.38

31 1.81
Ga
Gallium
69.72

32 2.01
Ge
Germanium
72.63

33 2.18
As
Arsenic
74.92

34 2.55
Se
Selenium
78.97

35 2.96
Br₂
Bromine
79.90

36 3.00
Kr
Krypton
83.80

5

37 0.82
Rb
Rubidium
85.47

38 0.95
Sr
Strontium
87.62

39 1.22
Y
Yttrium
88.91

40 1.33
Zr
Zirconium
91.22

41 1.6
Nb★
Niobium
92.91

42 2.16
Mo★
Molybdenum
95.95

43 1.9
Tc
Technetium
(98)

44 2.2
Ru★
Ruthenium
101.07

45 2.28
Rh★
Rhodium
102.91

46 2.20
Pd★★
Palladium
106.42

47 1.93
Ag★
Silver⁽¹⁺⁾
107.87

48 1.69
Cd
Cadmium
112.41

49 1.78
In
Indium
114.82

50 1.96
Sn
Tin
118.71

51 2.05
Sb
Antimony
121.76

52 2.1
Te
Tellurium
127.60

53 2.86
I₂
Iodine
126.90

54 2.60
Xe
Xenon
131.29

6

55 0.79
Cs
Caesium
132.91

56 0.89
Ba
Barium
137.33

57-71
La-Lu
Lanthanide

72 1.3
Hf
Hafnium
178.49

73 1.5
Ta
Tantalum
180.95

74 2.36
W
Tungsten
183.84

75 1.9
Re
Rhenium
186.21

76 2.2
Os
Osmium
190.23

77 2.20
Ir
Iridium
192.22

78 2.28
Pt★
Platinum
195.08

79 2.54
Au★
Gold
196.97

80 2.00
Hg
Mercury
200.59

81 1.62
Tl
Thallium
204.38

82 1.87
Pb
Lead
207.2

83 2.02
Bi
Bism

Polyatomic Ions		
I+		2-
(NH ₄) ⁺¹	ammonium	(CrO ₄) ⁻² (Cr ₂ O ₇) ⁻² (CO ₃) ⁻² (HPO ₄) ⁻²
I-		
(NO ₃) ⁻¹ (NO ₂) ⁻¹ (OH) ⁻¹ (HCO ₃) ⁻¹ (C ₂ H ₃ O ₂) ⁻¹ (ClO ₄) ⁻¹ (ClO ₃) ⁻¹ (ClO ₂) ⁻¹ (ClO) ⁻¹	nitrate nitrite hydroxide <u>bicarbonate</u> or hydrogen carbonate acetate perchlorate chlorate chlorite hypochlorite	manganate peroxide thiosulfate sulfate sulfite oxalate
(CN) ⁻¹ (SCN) ⁻¹ (HSO ₄) ⁻¹ (MnO ₄) ⁻¹ (H ₂ PO ₄) ⁻¹ (IO ₄) ⁻¹ (IO ₃) ⁻¹ (IO) ⁻¹	cyanide thiocyanate bisulfate or hydrogen sulfate permanganate dihydrogen phosphate periodate iodate hypoiodite	3- arsenate arsenite borate citrate phosphate or tribasic phosphate phosphite
(NH ₂) ⁻¹ (CHO ₂) ⁻¹	amide formate	4- silicate (ortho)

Atomic Ions		
+1		-1
Li ⁺¹ Na ⁺¹ K ⁺¹ Ag ⁺¹ Cu ⁺¹	Lithium Sodium Potassium Silver Copper (I)	Fluoride Bromide Chloride iodide
+2		-2
Mg ⁺² Ca ⁺² Ba ⁺² Zn ⁺² Cd ⁺² Hg ⁺² Hg ₂ ⁺² Cu ⁺² Pb ⁺² Fe ⁺² Ni ⁺² Mn ⁺² Sn ⁺²	Magnesium Calcium Barium Zinc Cadmium (II) Mercury (II) Mercury (I) Copper (II) Lead (II) Iron (II) Nickel (II) Manganese (II) Tin (II)	O ⁻² S ⁻² Oxide Sulfide
+3		-3
Al ⁺³ Fe ⁺³ Ni ⁺³	Aluminum Iron (III) Nickel (III)	Nitride Phosphide
+4		
Pb ⁺⁴ Si ⁺⁴ Sn ⁺⁴ Mn ⁺⁴	Lead (IV) Silicon (IV) Tin (IV) Manganese (IV)	